



TA1: The Mathematics of Agentic Communication Protocols

Agentic Platform

of agents: 2 (configurable per campaign)
types of agents: Math-reasoning orchestrator + SSM science validator
models: ChemBERTa-77M + Qwen2.5-Coder-0.5B-Instruct
tasks / domains: 2 NMR task families + 1 generalization lane (Hammett)
months to build (current) platform: 18

Proposed Mathematics/Systems/Information/Data Science Methods

Metrics

TA1 Rubric	Current results	Month 4 Target
% success rate	Terminal-bench: 13/13 (100%) Petri: 172/173 (99.42%) Real-patch: 5/5 (100%)	CDPTI >=95% recall; ACV_norm >=0.90
Governance overhead	Zero overhead (terminal-bench-core-0.1.1)	<=5% overhead on governed benchmark runs
IV&V bundle format	JSONL governance_events + latent_states.npz defined	Bundle locked at M1; phi-ablation complete by M3

Domain and Tasks

TA1 Field	Details
Selected Science Subdomain(s)	NMR spectroscopy (primary); Hammett equation (generalization)
1st family of scientific tasks	Task A: 1H NMR spectrum to molecular structure (BMRB+PDB)
2nd family of scientific tasks	Task B: 2D COSY to vicinal J-coupling connectivity (BMRB)
Small science models (SSMs)	ChemBERTa-77M + Qwen2.5-Coder-0.5B-Instruct



Proposed Metric #1 — Mathematical Evidence Emission (MEE)

Technical Area: TA1

Description: MEE measures the rate and verification quality of checkable mathematical artifacts emitted during inter-agent communication: formulas, derivation steps, citations, executable code, and numerical assertions.

Method for calculation: For each act, extract evidence artifacts and score each with a class-appropriate verifier (symbolic simplification, sandbox execution, dimensional checks, citation resolution). $MEE = \text{sum(scores)} / \# \text{ inter-agent messages}$. Report MEE_density, MEE_pass_rate, and artifact-class balance. IV&V inputs: governance_events.jsonl, evidence_artifacts.jsonl, task_manifests.json.

Progress against program goals: MATHBAC seeks science-discovery protocols that are understandable, not only successful. MEE distinguishes a correct black-box answer from a campaign that leaves a mathematical audit trail.

Supplement to current metrics: Program success, speed, and generalization are end-state measures. MEE is trace-level: did the protocol produce falsifiable scientific evidence?

Adoption argument: MEE gives IV&V a low-cost audit surface using the same evidence log needed to check math content. Vendor-neutral; computable from JSONL without SCBE source.

Literature: Minerva (arXiv:2206.14858), Polu/Sutskever (arXiv:2009.03393), AlphaGeometry (Nature 2024), LeanDojo (NeurIPS 2023).

Phase I targets: M3 — extractor and verifier calibration. M9 — MEE density/pass-rate on NMR task families. M13 — compare vs. Mixtral-8x7B baseline.



Proposed Metric #2 — Axiom Compliance Vector (ACV)

Technical Area: TA1

Description: ACV is a five-component vector in $[0,1]^5$ measuring per-act compliance with Unitarity, Locality, Causality, Symmetry, and Composition.

Method for calculation: Evaluate five predicates per act: information conservation (Unitarity), bounded influence radius (Locality), dependency-graph topological order (Causality), role-permutation invariance (Symmetry), pipeline schema continuity (Composition). Campaign ACV = mean predicate vector. $ACV_norm = (5 - ||1 - ACV||_1) / 5$. IV&V inputs: governance_events.jsonl, latent_states.npz, task_manifests.json.

Progress against program goals: TA1 seeks stability and convergence analysis. ACV measures whether structural conditions for operator well-posedness hold.

Supplement to current metrics: A protocol can be fast and correct while violating causality or composition. ACV catches structurally pathological protocols.

Adoption argument: Vendor-neutral — any performer can emit five predicate values from ordinary communication and latent-state logs. Provides Phase II teams a compatibility check before combining protocols.

Literature: von Neumann formalism, Pearl causality, Mac Lane composition, Goyal RIM.

Phase I targets: M3 — ACV field in event schema. M9 — mean $ACV_norm \geq 0.90$.



Proposed Metric #3 — Communication-Dynamics Phase-Transition Index (CDPTI)

Technical Area: TA1

Description: CDPTI measures signed regime changes in the multi-agent communication graph, detecting communication breakdown and coordinated adversarial convergence.

Method for calculation: Build time-windowed directed graph G_t . Edge (i,j) weighted by communication energy, signed by harmonic-wall evaluation: $w_{ij}(t) = \text{sign}(H_{ij}(t) - H_{\min}) * ||\log(S_{ij})||$. Trust-weighted directed Laplacian: $L_H(G_t) = D_{\text{out}}(W_H) - W_H$. Signed readout: $\text{CDPTI}(t) = \text{Re}(\lambda_2(L_H(G_t))) / \text{Re}(\lambda_2(L_H(G_{t0})))$. Negative signed-connectivity growth triggers escalation even if unsigned rises. IV&V inputs: governance_events.jsonl, latent_states.npz, protocol_graph_edges.jsonl.

Progress against program goals: PA line 443 — characterize communication dynamics. CDPTI identifies regime changes and their direction (safe vs. adversarial basin).

Supplement to current metrics: Outcome metrics don't show whether a campaign progressed smoothly or converged adversarially. CDPTI adds temporal/directional resolution.

Adoption argument: Directly addresses non-compositional safety (Spera 2026): conjunctive drift appears as signed graph motion rather than isolated local actions.

Literature: Newman spectra, Strogatz dynamics, Cohen-Steiner, Spera (arXiv:2603.15973).

Phase I targets: M3 instrument L_H . M6 live CDPTI. M13 $\geq 95\%$ recall, $\leq 5\%$ FPR.



Proposed Metric #4 — Protocol Information Signature (PIS)

Technical Area: TA1

Description: PIS is a fixed-dimensional protocol fingerprint summarizing the structural identity of a multi-agent protocol, independent of agent names and surface formatting.

Method for calculation: Compute mutual-information features, conditional-entropy graph features, role-transition features, latent-delta covariance summaries, and persistent-homology features. Project via released `pis_projection_matrix.json`. Compare with cosine distance and cluster persistence. IV&V inputs: `governance_events.jsonl`, `latent_states.npz`, `protocol_graph_edges.jsonl`, `pis_projection_matrix.json`.

Progress against program goals: TA1 deliverable C requires a protocol catalog. PIS is the catalog key: determines whether two protocols are structurally similar.

Supplement to current metrics: Success/speed do not identify whether a protocol is new, redundant, or transferable. PIS adds identity and similarity structure to the rubric.

Adoption argument: Detects benchmark gaming, supports cross-subdomain transfer, and enables catalog queries by structural similarity.

Literature: Cover/Thomas, Tishby bottleneck, Indyk/Motwani ANN, Charikar similarity estimation, persistent-homology stability.

Phase I targets: M6 — preliminary PIS feature extractor. M13 — catalog v1 with reference centroids and cluster-separation report.